Practice Exam A: Answer Key and Worked Solutions

Grade yourself honestly: points need correct units, sig figs and shown work. Red = where people usually lose points.

Problem 1: VSEPR / polarity (15)

(a) ClF_3: 7 + 3×7 = 28 e^-. Cl central, 3 F single-bonded, 2 lone pairs on Cl → AX_3E_2 (SN 5, trigonal-bipyramidal e^- geometry, LPs equatorial) → **T-shaped, F–Cl–F < 90°, POLAR**. Axial bond dipoles cancel; net dipole lies along the Cl–F(equatorial) axis, arrow pointing from F (δ^-) toward Cl (δ^+) (IUPAC).

(b) XeF_2: 8 + 14 = 22 e^-. AX_2E_3, 3 LPs equatorial → **linear, 180°, NON-POLAR** (bond dipoles cancel).

(c) PCl_3: 5 + 21 = 26 e^-. AX_3E → **trigonal pyramidal, Cl–P–Cl < 109.5° (~100°), POLAR**. Draw with 1 wedge + 1 dash + 1 line. Net dipole along the C_3 axis, arrow from the Cl_3 base (δ^-) toward P (δ^+).

Lone pairs on the F/Cl atoms must be drawn. Lone pairs on the central atom must be shown in the 3D drawing.

Problem 2: VB description of C_2H_4 (20)

Lewis: $H_2C{=}CH_2$ (12 e^-, no lone pairs). Each C has 3 neighbors, SN 3 → **sp^2 on both C, trigonal planar, ≈120° (H–C–H slightly < 120°, ~117°, because the C=C needs more space)**. Molecule is planar.

Axes: put the molecule in the **xy-plane** with C=C along x; z is perpendicular to the plane. sp^2 hybrids are made from 2s + $2p_x$ + 2p_y; **2p_z is left unhybridized** on each C.

- 4 × C–H: **σ**, C sp^2 + H 1s
- C=C: 1 **σ** (sp^2 + sp^2, head-on along x) + 1 **π** (2p_z + 2p_z, sideways; lobes above and below the molecular plane, nodal plane = molecular plane)

Total 5 σ + 1 π. π is weaker than σ. Sketch: two sp^2 lobes overlapping on the C–C axis, with two p_z dumbbells side by side forming the π cloud above and below.

Problem 3: MO of NO^+ / NO• / NO^- (20)

Valence e^-: NO^+ 10, NO 11, NO^- 12. Order (N_2-type): σ2s < σ*2s < π2p(×2) < σ2p < π*2p(×2) < σ*2p. (No g/u labels: heteronuclear molecules have no inversion center.)

- **NO^+**: $(\sigma 2s)^2(\sigma^* 2s)^2(\pi 2p)^4(\sigma 2p)^2$ → BO = ½(8−2) = **3, diamagnetic, 106.3 pm**
- **NO•**: …$(\sigma 2p)^2(\pi^* 2p)^1$ → BO = ½(8−3) = **2.5, paramagnetic (1 unpaired), 115.1 pm**
- **NO^-**: …$(\sigma 2p)^2(\pi^* 2p)^2$ (one e^- in each π*, Hund) → BO = ½(8−4) = **2, paramagnetic (2 unpaired), 125.8 pm**

Higher BO → shorter bond. Bonding MOs reside mostly on **O**: O is more electronegative, so its AOs are lower in energy, and a bonding MO resembles the AO closest to it in energy. (Antibonding MOs lie mostly on N.)

Problem 4: Methyl isocyanate (15)

e^-: 2×4 + 3×1 + 5 + 6 = 22. Structure **H_3C–N̈=C=Ö**: N has 1 LP, O has 2 LPs, **all formal charges = 0** (C: 4−0−4, N: 5−2−3, O: 6−4−2).

Hybridization: CH_3 carbon **sp^3**; N (2 atoms + 1 LP, SN 3) **sp^2**; central C (2 atoms, SN 2) **sp**; O (1 atom + 2 LP) **sp^2**.

Angles: C–N–C **≈ 120° (bent, slightly less because of the LP; the real value is ~140°)**; N–C–O **180° (linear)**.

$\mathscr{M}$ = 2(12.011) + 3(1.008) + 14.007 + 15.999 = **57.052 g/mol**.

Problem 5: Magnetite (10)

Assume 100 g: Fe 72.36/55.845 = 1.2957 mol; O 27.64/15.999 = 1.7276 mol. Divide by the smaller: Fe 1, O 1.3333 = 4/3 → ×3 → **Fe_3O_4**. Don't round 1.333 to 1!

Problem 6: Caffeine (15)

C 49.48/12.011 = 4.120; H 5.19/1.008 = 5.149; N 28.85/14.007 = 2.060; O 16.48/15.999 = 1.030. ÷1.030 → 4.00 : 5.00 : 2.00 : 1.00 → empirical **$C_4H_5N_2O$** (97.10 g/mol). k = 194/97.10 = 2.00 → molecular **$C_8H_{10}N_4O_2$**.

Problem 7: Blast furnace (20)

(a) **$Fe_2O_3 + 3\,CO \rightarrow 2\,Fe + 3\,CO_2$**

(b) Fe_2O_3: 50.0/159.687 = 0.3131 mol (÷1 = 0.3131). CO: 25.0/28.010 = 0.8925 mol (÷3 = 0.2975). Smaller → **CO is limiting**.

(c) Fe = 0.8925 × (2/3) × 55.845 = **33.2 g Fe (theoretical)**. Fe_2O_3 used = 0.8925/3 = 0.2975 mol × 159.687 = 47.51 g → left = 50.0 − 47.5 = **2.5 g Fe_2O_3 excess**.

(d) %-yield = 28.9/33.23 × 100% = **87.0%**.

Redox note: Fe +3→0 (reduced), C +2→+4 (oxidized).

Problem 8: Dilution (10)

$V_1 = c_2V_2/c_1$ = (0.0750 M)(250. mL)/(6.00 M) = **3.13 mL**. Measure it with a pipet, add it to a 250-mL **volumetric flask** already partly filled with DI water (**acid into water**), fill to the mark with DI water, then stopper and invert to mix.

Problem 9: Molal solution from a hydrate (15)

n($CuSO_4$) = 0.2000 mol/kg × 0.5000 kg = 0.1000 mol. $\mathcal{M}(CuSO_4 \cdot 5H_2O)$ = 159.60 + 5(18.015) = 249.68 g/mol → weigh **24.97 g of the pentahydrate**.

The hydrate brings 5 × 0.1000 mol × 18.015 g/mol = 9.008 g of water, which counts as solvent. So weigh out only **491.0 g DI water** (500.0 − 9.0). Weigh both on an analytical balance, combine, stir until dissolved, and cover to prevent evaporation.

Molality is based only on **masses** (more accurate than volumes) and is **temperature-independent**; molarity depends on solution volume, which changes with T.

Classic mistake: using 159.60 g/mol (anhydrous) or forgetting the hydrate water.

Problem 10: $NiCl_2$ + NaOH (15)

(a) $NiCl_2(aq)$: $Ni^{2+}(aq)$, $Cl^-(aq)$, $H_2O(l)$. NaOH(aq): $Na^+(aq)$, $OH^-(aq)$, $H_2O(l)$.
(b) **Precipitation** (transition-metal hydroxides are insoluble; NaCl is soluble).
(c) **$Ni^{2+}(aq) + 2\ OH^-(aq) \rightarrow Ni(OH)_2(s)$** (green). Spectators: Na^+, Cl^-.
(d) Ni^{2+} = 0.100 L × 0.0200 M = 2.00×10^{-3} mol; OH^- = 3.00×10^{-3} mol, but 4.00×10^{-3} mol would be needed → **OH^- (NaOH) limiting**. $Ni(OH)_2$ = 1.50×10^{-3} mol × 92.707 = **0.139 g**.

Problem 11: Titration (15)

$H_2SO_4 + 2\ NaOH \rightarrow Na_2SO_4 + 2\ H_2O$. n(NaOH) = 0.03142 L × 0.1500 M = 4.713×10^{-3} mol → ÷2 = 2.3565×10^{-3} mol H_2SO_4 → ÷ 0.02500 L = **0.09426 M**.

Equivalence point = stoichiometrically equivalent amount of base added. End point = the indicator changes color (a tiny excess past equivalence). Diprotic acid: 2:1 ratio!

Problem 12: Gas collected over water (15)

P(H_2) = 755 − 23.8 = 731.2 mm-Hg = 0.9621 atm. T = 298.15 K, V = 0.245 L.
n = PV/RT = (0.9621)(0.245)/(0.082057 × 298.15) = 9.635×10^{-3} mol H_2 = mol Zn (1:1) → × 65.38 = **0.630 g Zn**.

Subtract the water vapor pressure; mL → L; °C → K.

Problem 13: Unknown gas (15)

(a) $\mathcal{M}$ = mRT/(PV) = (1.28)(0.082057)(300.15)/[(0.987)(0.500)] = **63.9 g/mol → SO_2** (64.06).
(b) d = m/V = 1.28 g/0.500 L = **2.56 g/L** (check: P$\mathcal{M}$/RT = 2.56 ✓).
(c) rate(SO_2)/rate(O_2) = √(32.00/64.06) = **0.707**: SO_2 effuses at 0.707× the rate of O_2 (O_2 is 1.41× faster).
(d) O=S–O ↔ O–S=O resonance (or O=S=O with an expanded octet), 1 LP on S → AX_2E → **bent, ≈119°, polar**, S sp^2.

Problem 14: Pyrophorics (10)

(a) Any two: t-BuLi is **pyrophoric** (ignites spontaneously on contact with air → fire/severe burns); pentane is **highly flammable**; reacts violently with water/moisture; corrosive to skin; syringe plunger can separate and spill (Sangji, UCLA 2009).
(b) ≥ 50 mL → **cannula (double-tipped needle) transfer under inert gas** (N_2/Ar) through rubber septa, not a syringe.
(c) **Flame-resistant lab coat, safety goggles + face shield, gloves** (e.g. nitrile, ideally under fire-resistant gloves).
(d) **SDS Section 8** (Exposure controls / personal protection).

Problem 15: Redox (10)

Cu(s): **0**. HNO_3: H +1, **N +5**, O −2. $Cu(NO_3)_2$: **Cu +2**, N +5, O −2. NO_2: **N +4**, O −2. H_2O: H +1, O −2.

Cu oxidized (0→+2) = reducing agent; N reduced (+5→+4) = HNO_3 is the oxidizing agent. (Only 2 of the 4 N atoms are reduced; the other 2 stay +5 in nitrate.)

Problem 16: Kinetic-molecular theory (10)

(a) Average kinetic energy depends only on T → **all equal**. u_rms ∝ 1/√$\mathcal{M}$ → **$SO_2 < N_2 < He$**.
(b) u_rms = √(3RT/$\mathcal{M}$) = √(3 × 8.314 × 298.15 / 0.028014 kg/mol) = **515 m/s**. $\mathcal{M}$ must be in kg/mol!
(c) **High P and low T**: the particles are close together, so their own volume is no longer negligible, and they move slowly, so intermolecular attractions matter. Both break the ideal-gas assumptions (point particles, no forces).